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PROJET : COMPANY NETWORK DESIGN &
IMPLEMENTATION

ANNEES SCOLAIRE : 2025 /2026

PROFESSEUR : Mme KISSI

Nous voulons vous signaler un petit problème de son que nous avons rencontré avec la vidéo . Nous en sommes profondément désolé.

Introduction

Dans le cadre de l'agrandissement d'un centre de support de salle de marché employant 600 personnes, un nouveau bâtiment a été identifié pour accueillir les activités de l'entreprise. Ce bâtiment ne disposant d'aucune infrastructure réseau existante, il est nécessaire de concevoir et de mettre en œuvre un nouveau réseau informatique fiable, sécurisé et évolutif .

L'objectif de ce projet est de proposer une conception logique du réseau répondant aux besoins actuels de l'entreprise tout en permettant une évolution future. La solution est conçue et simulée à l'aide de Cisco Packet Tracer, en respectant les bonnes pratiques de conception réseau.

PLAN DU PROJET :

- 1 Dessiner le schéma (papier ou Packet Tracer)
- 2 Créer les VLANs
- 3 Faire le subnetting
- 4 Configurer les multilayers
- 5 Ajouter DHCP
- 6 OSPF
- 7 NAT / PAT
- 8 Sécurité
- 9 Tests

Architecture générale du réseau

Le réseau proposé est basé sur un modèle hiérarchique à trois couches, recommandé par Cisco pour les réseaux d'entreprise :

Couche cœur (Core Layer) : assurée par deux routeurs pour garantir la redondance et la connexion à Internet.

Couche distribution : assurée par deux commutateurs multilayers (Layer 3) responsables du routage inter-VLAN.

Couche accès : assurée par des commutateurs d'accès connectant les utilisateurs et les points d'accès sans fil.

Cette architecture permet une meilleure performance, une gestion simplifiée et une haute disponibilité du réseau.

Organisation des départements

Le bâtiment comporte trois étages avec les départements suivants :

Premier étage :

Ventes et Marketing (120 utilisateurs)

Ressources humaines et Logistique (120 utilisateurs)

Deuxième étage

Finances et Comptabilité (120 utilisateurs)

Administration et Relations publiques (120 utilisateurs)

Troisième étage

ICT (120 utilisateurs)

Salle des serveurs (12 équipements)

Chaque département est isolé logiquement grâce à l'utilisation de VLANs.

Imagine le bâtiment comme **un grand immeuble de bureaux** :

- 3 étages
- 6 départements (120 personnes chacun)
- 1 salle serveurs (très importante)
- Tout le monde doit :
 - Avoir Internet
 - Communiquer entre départements
 - Être sécurisé
 - Continuer à fonctionner même si un équipement tombe en panne

👉 **Ton rôle** : concevoir **le cerveau du réseau**.

On te demande un modèle hiérarchique Cisco, qui a 3 couches :



◆ Pourquoi ce modèle ?

Parce que :

- C'est plus stable
- C'est évolutif
- C'est plus facile à dépanner

VLAN et plan d'adressage IP

Chaque département est placé dans un VLAN distinct et dispose de son propre sous-réseau IP afin d'améliorer la sécurité et l'organisation du réseau. Le réseau privé de base utilisé est 172.16.1.0, qui a été subnetté comme suit

Département	VLAN	Réseau IP	
Ventes / Marketing	VLAN 10	172.16.1.0/25	
RH / Logistique	VLAN 20	172.16.1.128/25	
Finances / Comptabilité	VLAN 30	172.16.2.0/25	
Administration / RP	VLAN 40	172.16.2.128/25	
ICT	VLAN 50	172.16.3.0/25	
Salle des serveurs	VLAN 60	172.16.3.128/28	

Chaque VLAN dispose d'une interface virtuelle (SVI) sur les commutateurs multilayers, servant de **passerelle par défaut** pour les utilisateurs.

Wi-Fi par département

Chaque département doit avoir :

Un Access Point

Connecté à son VLAN

Les utilisateurs Wi-Fi sont dans le même réseau que les PCs

➡ Même VLAN = même sous-réseau = communication facile

Routing inter-VLAN

Le routage entre les différents VLANs est assuré par les commutateurs multilayers, qui combinent les fonctions de commutation et de routage. Cette configuration permet aux différents départements de communiquer entre eux de manière efficace.

Base Network: 172.16.1.0

First Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Sales & Marketing	172.16.1.0	255.255.255.128/25	172.16.1.1 to 172.16.1.126	172.16.1.127
HR and Logistics	172.16.1.128	255.255.255.128/25	172.16.1.129 to 172.16.1.254	172.16.1.255

Second Floor

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
Finance & Accounts	172.16.2.0	255.255.255.128/25	172.16.2.1 to 172.16.2.126	172.16.2.127
Admin & Public Relations	172.16.2.128	255.255.255.128/25	172.16.2.129 to 172.16.2.254	172.16.2.255

Department	Network Address	Subnet Mask	Host Address Range	Broadcast Address
ICT	172.16.3.0	255.255.255.128/25	172.16.3.1 to 172.16.3.126	172.16.3.127
Server Room	172.16.3.128	255.255.255.240/28	172.16.3.129 to 172.16.3.142	172.16.3.143

Between the Routers and Layer-3 Switches

No.	Network Address	Subnet Mask	Host Address Range	Broadcast Address
R1- MLSW1	172.16.3.144	255.255.255.252	172.16.3.145 to 172.16.3.146	172.16.3.147
R1- MLSW2	172.16.3.148	255.255.255.252	172.16.3.149 to 172.16.3.150	172.16.3.151
R2- MLSW1	172.16.3.152	255.255.255.252	172.16.3.153 to 172.16.3.154	172.16.3.155
R2- MLSW2	172.16.3.156	255.255.255.252	172.16.3.157 to 172.16.3.158	172.16.3.159

Between the Routers and Layer-3 Switches

Public IP addresses 195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30 and 195.136.17.12/30

AJOUT DE DHCP

OSPF

&

OSPF = protocole de routage dynamique

Pourquoi OSPF ?

- Découvre automatiquement les routes
- Très utilisé en entreprise
- Fonctionne avec la redondance

👉 OSPF sera configuré :

- Sur les routeurs
- Sur les commutateurs multilayers

PAT (NAT OVERLOAD) / ACL

Les PCs ont des IP **privées** (172.16.x.x)

Internet utilise des IP **publiques**

👉 **PAT :**

- Tous les utilisateurs partagent l'IP publique du routeur
- Le routeur fait la traduction

ACL = dit **quelles adresses peuvent sortir vers Internet**

SÉCURITÉ

🔒 12. Sécurité – Finances

Pour le VLAN Finances :

- Port-security
- 1 seul PC par port
- Sticky MAC
- Violation = shutdown

➡ Si quelqu'un branche un autre PC → port bloqué

TEST

🔑 13. SSH (accès distant sécurisé)

Sur :

- Routeurs
- Commutateurs multilayers

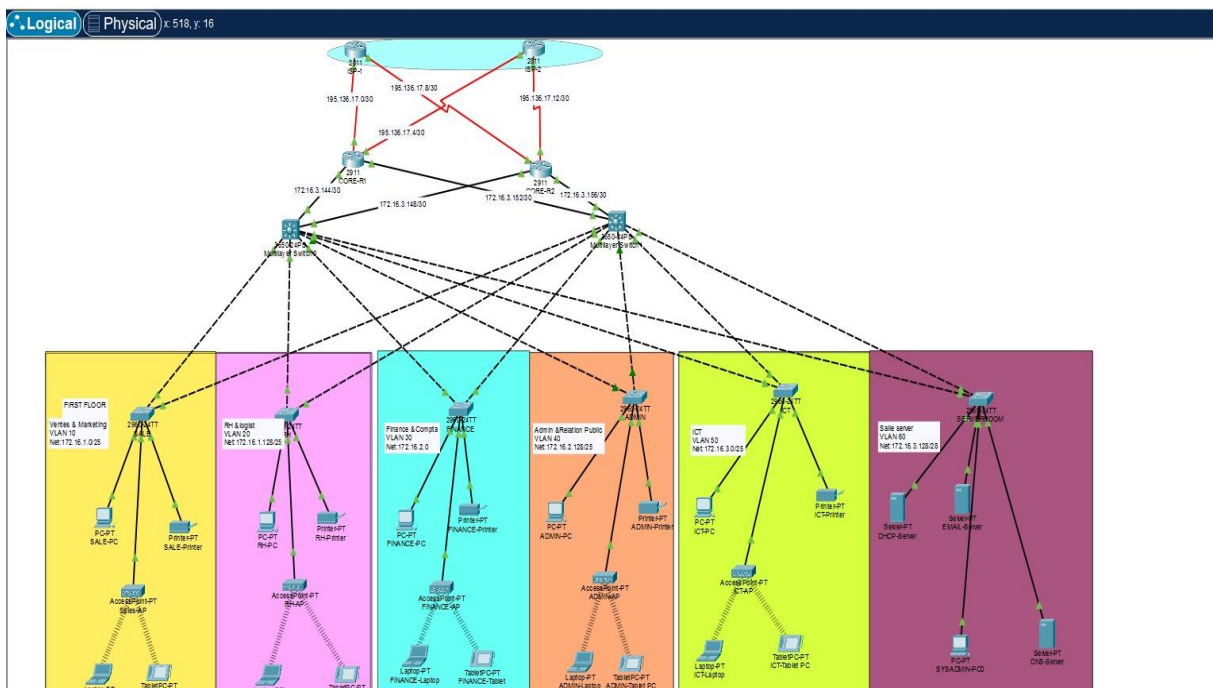
👉 Pas de Telnet (non sécurisé)

🧪 14. Tests finaux

- ✓ Ping entre VLANs
- ✓ Ping vers Internet
- ✓ DHCP fonctionne
- ✓ Redondance OK
- ✓ Sécurité ports OK

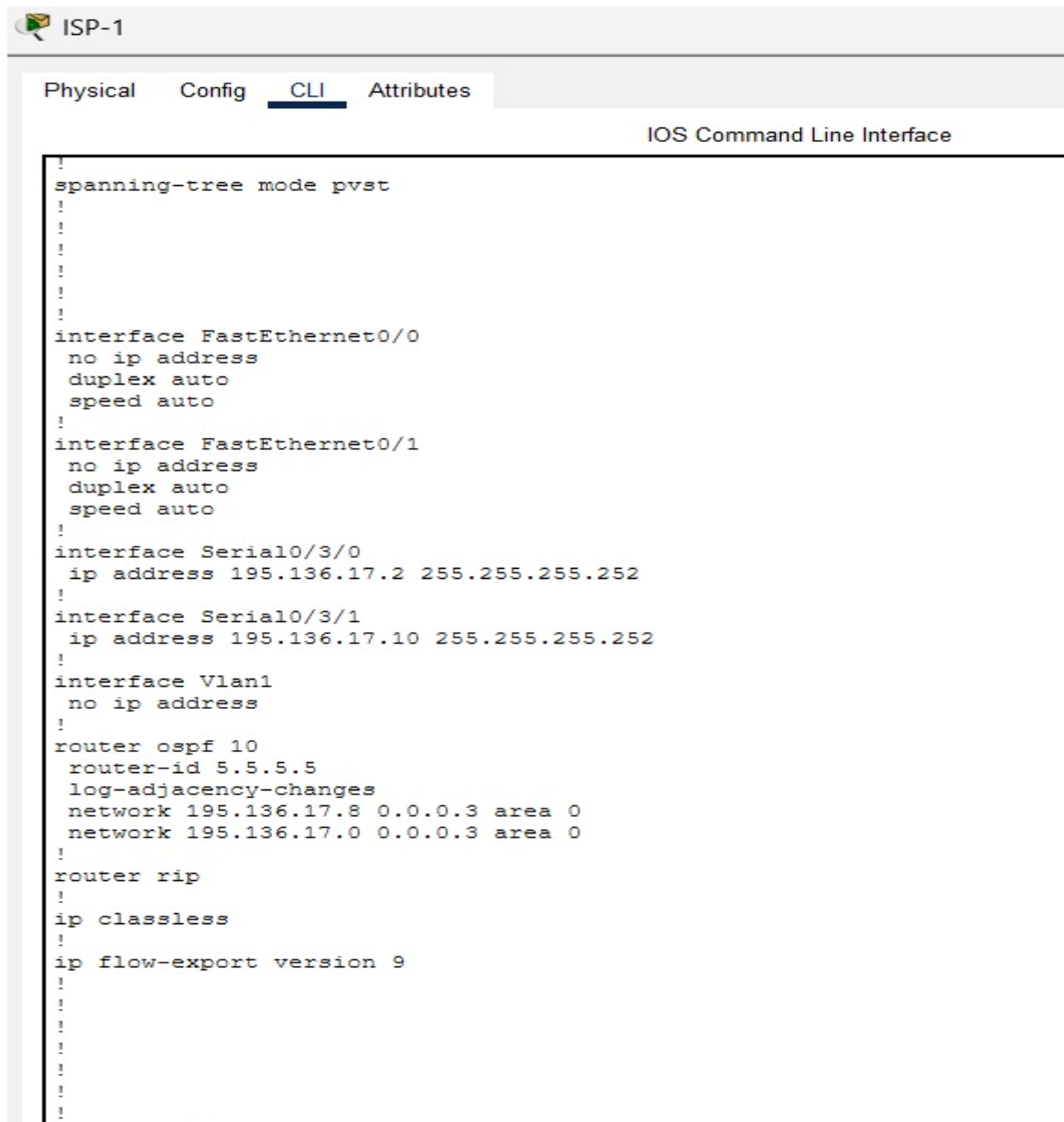
Maintenant nous allons passer à la réalisation :

Topologie du projet



Les configurations

Commençons par **ISP1** :



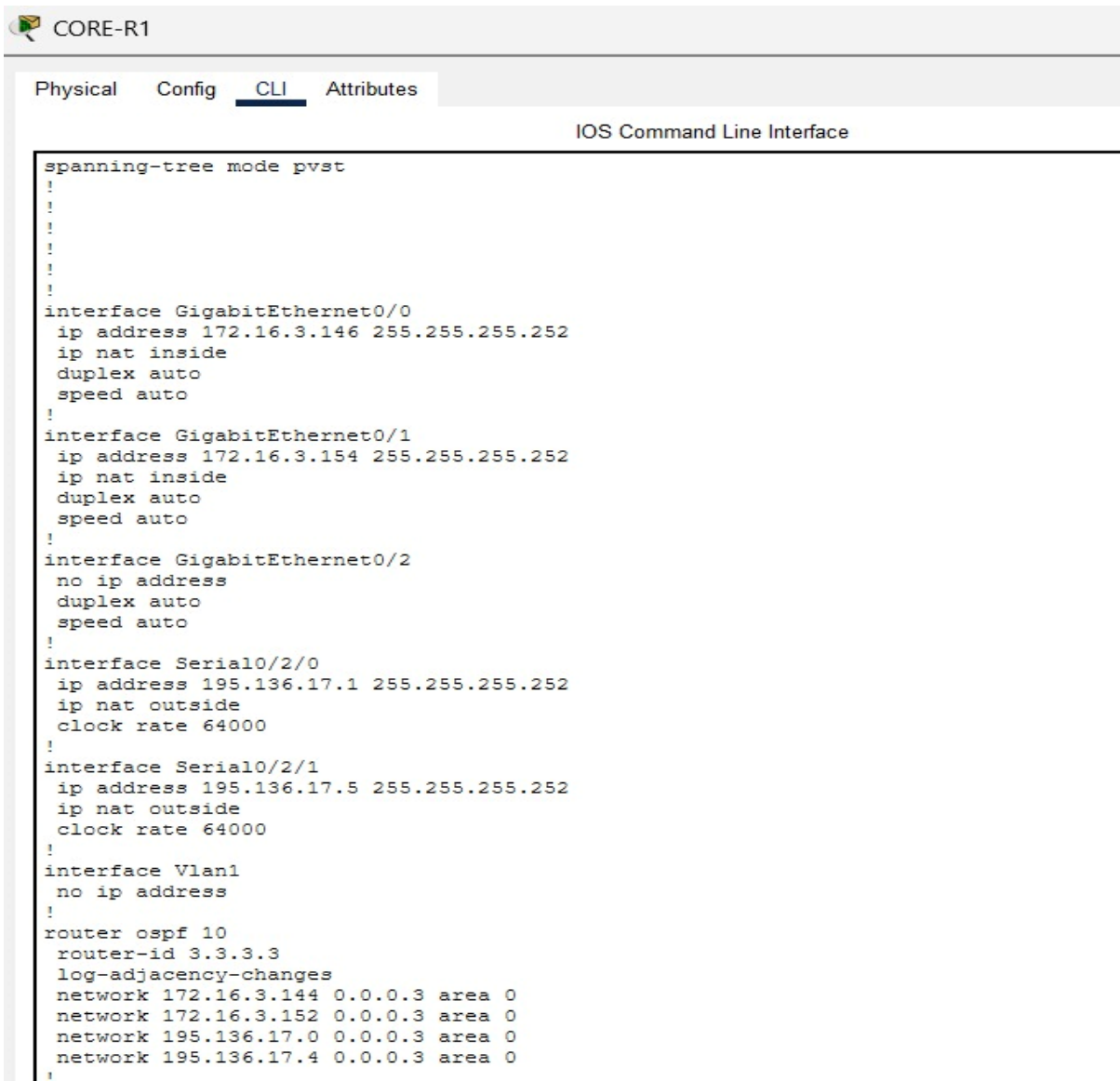
```
ISP-1
Physical Config CLI Attributes
IOS Command Line Interface

!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet0/1
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  ip address 195.136.17.2 255.255.255.252
!
interface Serial0/3/1
  ip address 195.136.17.10 255.255.255.252
!
interface Vlan1
  no ip address
!
router ospf 10
  router-id 5.5.5.5
  log-adjacency-changes
  network 195.136.17.8 0.0.0.3 area 0
  network 195.136.17.0 0.0.0.3 area 0
!
router rip
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
```

ISP2 :

```
!
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet0/1
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  ip address 195.136.17.6 255.255.255.252
!
interface Serial0/3/1
  ip address 195.136.17.14 255.255.255.252
!
interface Vlan1
  no ip address
!
router ospf 10
  router-id 5.5.5.5
  log-adjacency-changes
  network 195.136.17.4 0.0.0.3 area 0
  network 195.136.17.12 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
```

ROUTEUR 1 OU CORE 1 :



```
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 172.16.3.146 255.255.255.252
ip nat inside
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 172.16.3.154 255.255.255.252
ip nat inside
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
!
interface Serial0/2/0
ip address 195.136.17.1 255.255.255.252
ip nat outside
clock rate 64000
!
interface Serial0/2/1
ip address 195.136.17.5 255.255.255.252
ip nat outside
clock rate 64000
!
interface Vlan1
no ip address
!
router ospf 10
router-id 3.3.3.3
log-adjacency-changes
network 172.16.3.144 0.0.0.3 area 0
network 172.16.3.152 0.0.0.3 area 0
network 195.136.17.0 0.0.0.3 area 0
network 195.136.17.4 0.0.0.3 area 0
!
```

La commande **ip nat inside ou outside** permet au routeur de savoir d'où vient le trafic et vers où il va . **ip nat inside** marque l'interface connecté au réseau local (@ip privée) et **ip nat outside** marque l'interface connecté vers l'extérieur (@ip public)

Access-list 1 permet jouer un rôle de filtre . Elle définit qui a le droit d'entrée ou sortir du réseau .

```
!  
interface Vlan1  
  no ip address  
!  
router ospf 10  
  router-id 3.3.3.3  
  log-adjacency-changes  
  network 172.16.3.144 0.0.0.3 area 0  
  network 172.16.3.152 0.0.0.3 area 0  
  network 195.136.17.0 0.0.0.3 area 0  
  network 195.136.17.4 0.0.0.3 area 0  
!  
ip nat inside source list 1 interface Serial0/2/1 overload  
ip classless  
ip route 0.0.0.0 0.0.0.0 Serial0/2/0  
ip route 0.0.0.0 0.0.0.0 Serial0/2/1 70  
!  
ip flow-export version 9  
!  
!  
access-list 1 permit 172.16.1.0 0.0.0.127  
access-list 1 permit 172.16.1.128 0.0.0.127  
access-list 1 permit 172.16.2.0 0.0.0.127  
access-list 1 permit 172.16.2.128 0.0.0.127  
access-list 1 permit 172.16.3.0 0.0.0.127  
access-list 1 permit 172.16.3.128 0.0.0.15  
!  
!  
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
  login  
!  
!  
!  
end
```

ROUTEUR 2 OU CORE 2 :

```
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 172.16.3.150 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 172.16.3.158 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
!
interface Serial0/2/0
 ip address 195.136.17.9 255.255.255.252
 ip nat outside
 clock rate 64000
!
interface Serial0/2/1
 ip address 195.136.17.13 255.255.255.252
 ip nat outside
 clock rate 64000
!
interface Vlan1
 no ip address
!
router ospf 10
 router-id 4.4.4.4
 log-adjacency-changes
 network 172.16.3.148 0.0.0.3 area 0
 network 172.16.3.156 0.0.0.3 area 0
 network 195.136.17.8 0.0.0.3 area 0
 network 195.136.17.12 0.0.0.3 area 0
!
```

```
!  
ip nat inside source list 1 interface Serial0/2/1 overload  
ip classless  
ip route 0.0.0.0 0.0.0.0 Serial0/2/0  
ip route 0.0.0.0 0.0.0.0 Serial0/2/1 70  
!  
ip flow-export version 9  
!  
!  
access-list 1 permit 172.16.1.0 0.0.0.127  
access-list 1 permit 172.16.1.128 0.0.0.127  
access-list 1 permit 172.16.2.0 0.0.0.127  
access-list 1 permit 172.16.2.128 0.0.0.127  
access-list 1 permit 172.16.3.0 0.0.0.127  
access-list 1 permit 172.16.3.128 0.0.0.15  
!  
!  
!  
!  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
  login  
!  
!  
!  
end  
  
CORE-R2#
```

MULTILAYER 1 :

```

spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet1/0/1
 no switchport
 ip address 172.16.3.145 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/2
 no switchport
 ip address 172.16.3.149 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet1/0/3
 switchport mode trunk
!
interface GigabitEthernet1/0/4
 switchport mode trunk
!
interface GigabitEthernet1/0/5
 switchport mode trunk
!
interface GigabitEthernet1/0/6
 switchport mode trunk
!
interface GigabitEthernet1/0/7
 switchport mode trunk
!
interface GigabitEthernet1/0/8
 switchport mode trunk
!

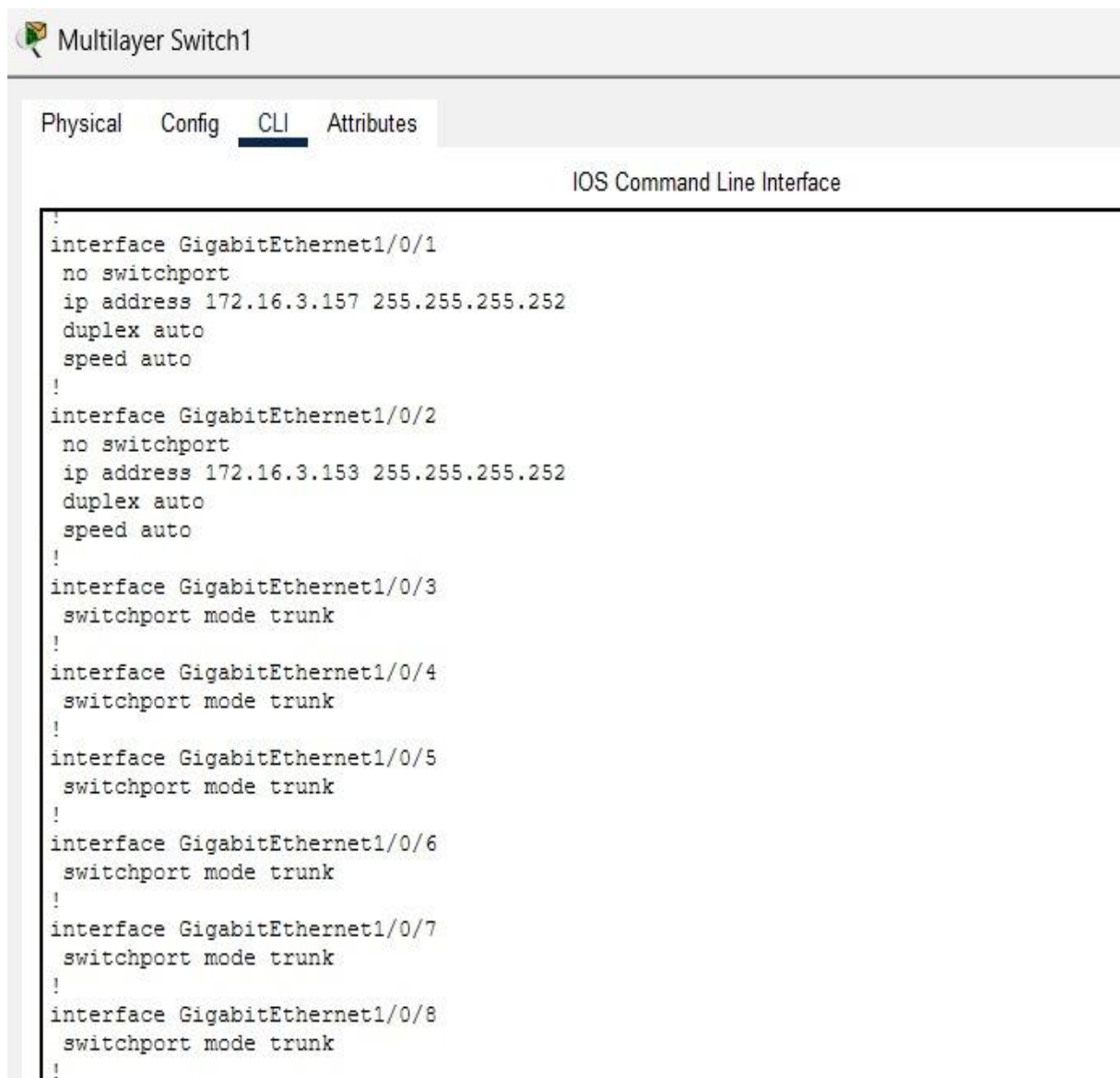
```

```
interface Vlan1
no ip address
!
interface Vlan10
mac-address 0006.2a7d.c801
ip address 172.16.1.1 255.255.255.128
ip helper-address 172.16.3.130
!
interface Vlan20
mac-address 0006.2a7d.c802
ip address 172.16.1.129 255.255.255.128
ip helper-address 172.16.3.130
!
interface Vlan30
mac-address 0006.2a7d.c803
ip address 172.16.2.1 255.255.255.128
ip helper-address 172.16.3.130
!
interface Vlan40
mac-address 0006.2a7d.c804
ip address 172.16.2.129 255.255.255.128
ip helper-address 172.16.3.130
!
interface Vlan50
mac-address 0006.2a7d.c805
ip address 172.16.3.1 255.255.255.128
ip helper-address 172.16.3.130
!
interface Vlan60
mac-address 0006.2a7d.c806
ip address 172.16.3.129 255.255.255.240
ip helper-address 172.16.3.130
!
router ospf 10
router-id 2.2.2.2
log-adjacency-changes
network 172.16.1.0 0.0.0.127 area 0
network 172.16.2.128 0.0.0.127 area 0
network 172.16.1.128 0.0.0.127 area 0
network 172.16.2.0 0.0.0.127 area 0
network 172.16.3.0 0.0.0.127 area 0
network 172.16.3.128 0.0.0.127 area 0
network 172.16.3.128 0.0.0.15 area 0
network 172.16.3.144 0.0.0.3 area 0
network 172.16.3.148 0.0.0.3 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 GigabitEthernet1/0/1
ip route 0.0.0.0 0.0.0.0 GigabitEthernet1/0/2 70
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
end

MTL-SW1#
```


Ip helper-address nous permet de rediriger les demandes d' @ip des clients vers le serveur DHCP

MULTILAYER 2 :



The screenshot shows a network management interface for a device named "Multilayer Switch1". The interface has four tabs: "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected and highlighted with a blue underline. Below the tabs, the text "IOS Command Line Interface" is displayed. The main area shows a series of configuration commands for various GigabitEthernet interfaces, separated by exclamation marks (!). The commands are as follows:

```
!
interface GigabitEthernet1/0/1
  no switchport
  ip address 172.16.3.157 255.255.255.252
  duplex auto
  speed auto
!
interface GigabitEthernet1/0/2
  no switchport
  ip address 172.16.3.153 255.255.255.252
  duplex auto
  speed auto
!
interface GigabitEthernet1/0/3
  switchport mode trunk
!
interface GigabitEthernet1/0/4
  switchport mode trunk
!
interface GigabitEthernet1/0/5
  switchport mode trunk
!
interface GigabitEthernet1/0/6
  switchport mode trunk
!
interface GigabitEthernet1/0/7
  switchport mode trunk
!
interface GigabitEthernet1/0/8
  switchport mode trunk
!
```

```

!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan10
  mac-address 0002.1600.cc01
  ip address 172.16.1.1 255.255.255.128
  ip helper-address 172.16.3.130
!
interface Vlan20
  mac-address 0002.1600.cc02
  ip address 172.16.1.129 255.255.255.128
  ip helper-address 172.16.3.130
!
interface Vlan30
  mac-address 0002.1600.cc03
  ip address 172.16.2.1 255.255.255.128
  ip helper-address 172.16.3.130
!
interface Vlan40
  mac-address 0002.1600.cc04
  ip address 172.16.2.129 255.255.255.128
  ip helper-address 172.16.3.130
!
interface Vlan50
  mac-address 0002.1600.cc05
  ip address 172.16.3.1 255.255.255.128
  ip helper-address 172.16.3.130
!
interface Vlan60
  mac-address 0002.1600.cc06
  ip address 172.16.3.129 255.255.255.240
  ip helper-address 172.16.3.130
!
router ospf 10
  router-id 1.1.1.1
  log-adjacency-changes
  network 172.16.1.0 0.0.0.127 area 0
  network 172.16.2.128 0.0.0.127 area 0
  network 172.16.1.128 0.0.0.127 area 0
  network 172.16.2.0 0.0.0.127 area 0
  network 172.16.3.0 0.0.0.127 area 0
  network 172.16.3.128 0.0.0.127 area 0
  network 172.16.3.128 0.0.0.15 area 0
  network 172.16.3.152 0.0.0.3 area 0
  network 172.16.3.156 0.0.0.3 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 GigabitEthernet1/0/1
ip route 0.0.0.0 0.0.0.0 GigabitEthernet1/0/2 70
!
ip flow-export version 9
!

```

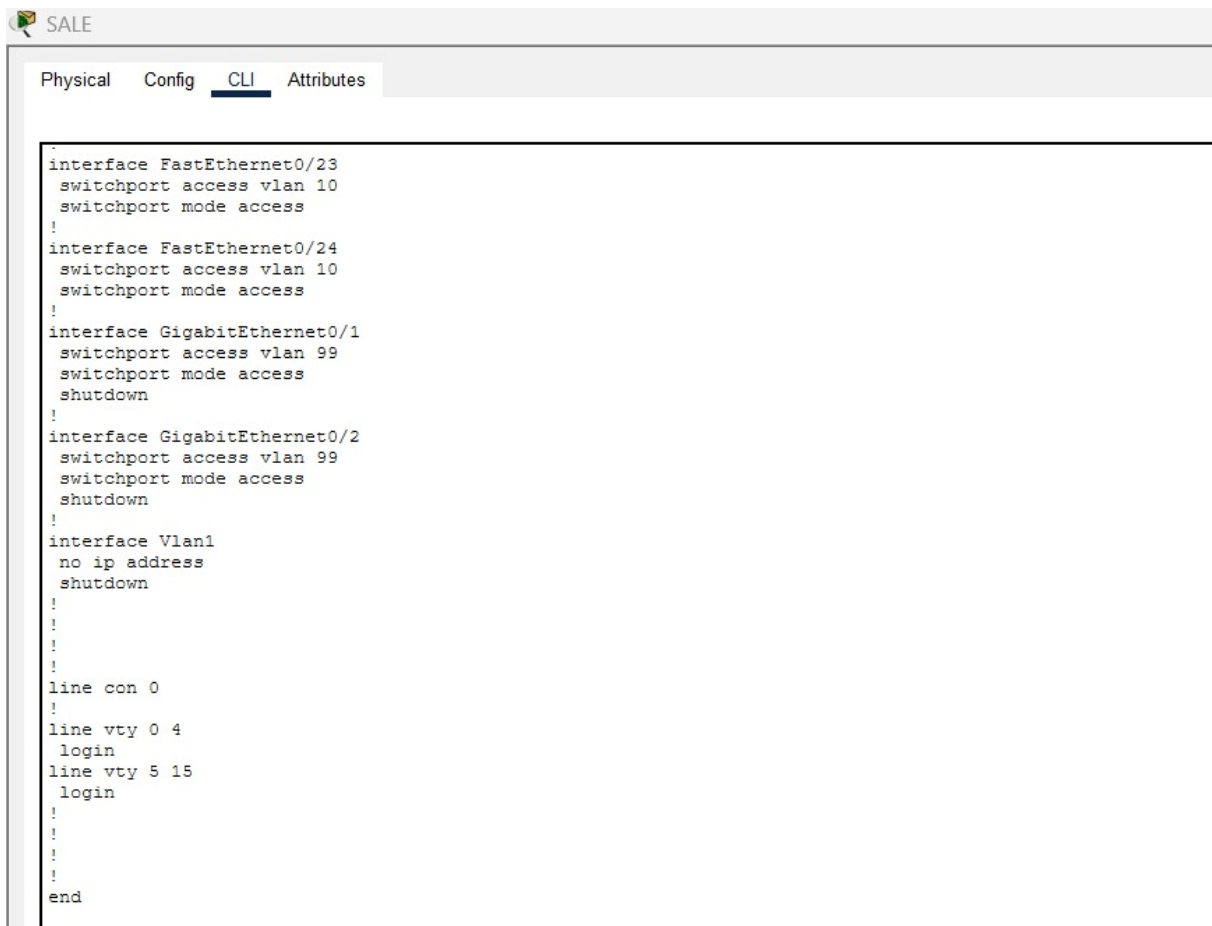
SALE & Marketing:

```
Sales#sh run
Building configuration...

Current configuration : 2369 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Sales
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 10
 switchport mode access
!
```

```
!
interface FastEthernet0/10
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/12
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/13
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/14
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/15
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/16
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/17
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/18
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/19
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/20
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/21
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/22
  switchport access vlan 10
  switchport mode access
!
```

Les ports fastEthernet 1 et 2 sont en mode trunk et tout le reste est mis en mode access vlan 10 pour permettre aux utilisateurs de ce vlan de pouvoir accéder aux réseaux . les gigabitEthernet sont reliés au vlan 99 de management .



HR & LOGISTICS :

```
HR>en
HR#sh run
Building configuration...

Current configuration : 2366 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname HR
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
,
```

```
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/14
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/15
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/16
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/17
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/18
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/19
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/20
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/21
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/22
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/23
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/24
 switchport access vlan 20
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 99
 switchport mode access
 shutdown
!
interface GigabitEthernet0/2
```

```
!
interface GigabitEthernet0/2
 switchport access vlan 99
 switchport mode access
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end
```


FINANCE & ACCOUNT :

 FINANCE

Physical Config CLI Attributes

```
FINANCE-SW>en
FINANCE-SW#sh run
Building configuration...

Current configuration : 3386 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname FINANCE-SW
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
    switchport mode trunk
!
interface FastEthernet0/2
    switchport mode trunk
!
interface FastEthernet0/3
    switchport access vlan 30
    switchport mode access
    switchport port-security mac-address sticky
!
interface FastEthernet0/4
    switchport access vlan 30
    switchport mode access
    switchport port-security mac-address sticky
!
interface FastEthernet0/5
    switchport access vlan 30
    switchport mode access
    switchport port-security mac-address sticky
!
interface FastEthernet0/6
    switchport access vlan 30
    switchport mode access
    switchport port-security mac-address sticky
!
interface FastEthernet0/7
    switchport access vlan 30
    switchport mode access
    switchport port-security mac-address sticky
!
```


FINANCE

Physical Config CLI Attributes

```
!
interface FastEthernet0/8
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/9
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/10
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/11
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/12
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/13
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/14
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/15
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/16
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/17
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
```

La commande port-security mac-address sticky nous permet securiser les ports configurées.

```

!
interface FastEthernet0/18
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/19
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/20
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/21
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/22
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/23
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface FastEthernet0/24
  switchport access vlan 30
  switchport mode access
  switchport port-security mac-address sticky
!
interface GigabitEthernet0/1
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface GigabitEthernet0/2
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!

```

ADMIN & PUBLIC RELATIONS :

```
ADMIN#sh run
Building configuration...

Current configuration : 2369 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ADMIN
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 40
 switchport mode access
!
```

```
interface FastEthernet0/10
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/12
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/13
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/14
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/15
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/16
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/17
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/18
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/19
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/20
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/21
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/22
  switchport access vlan 40
  switchport mode access
!
```

Physical Config CLI Attributes

```
!
interface FastEthernet0/23
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/24
  switchport access vlan 40
  switchport mode access
!
interface GigabitEthernet0/1
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface GigabitEthernet0/2
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
!
!
!
line con 0
!
line vty 0 4
  login
line vty 5 15
  login
!
!
!
!
end
```

ICT :

```

ICT>en
ICT#sh run
Building configuration...

Current configuration : 2367 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ICT
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
    switchport mode trunk
!
interface FastEthernet0/2
    switchport mode trunk
!
interface FastEthernet0/3
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/4
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/5
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/6
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/7
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/8
    switchport access vlan 50
    switchport mode access
!
interface FastEthernet0/9
    switchport access vlan 50
    switchport mode access

```



```

interface FastEthernet0/10
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/12
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/13
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/14
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/15
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/16
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/17
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/18
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/19
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/20
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/21
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/22
  switchport access vlan 50
  switchport mode access

```

```
!  
interface FastEthernet0/23  
  switchport access vlan 50  
  switchport mode access  
!  
interface FastEthernet0/24  
  switchport access vlan 50  
  switchport mode access  
!  
interface GigabitEthernet0/1  
  switchport access vlan 99  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet0/2  
  switchport access vlan 99  
  switchport mode access  
  shutdown  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
!  
!  
!  
line con 0  
!  
line vty 0 4  
  login  
line vty 5 15  
  login  
!  
!  
!  
!  
end
```

SERVER ROOM :

Physical Config CLI Attributes

```
ROOM-SERVER#sh run
Building configuration...

Current configuration : 2375 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname ROOM-SERVER
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport mode trunk
!
interface FastEthernet0/2
  switchport mode trunk
!
interface FastEthernet0/3
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/7
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/8
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/9
  switchport access vlan 60
  switchport mode access
!
```

```
interface FastEthernet0/10
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/11
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/12
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/13
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/14
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/15
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/16
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/17
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/18
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/19
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/20
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/21
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/22
  switchport access vlan 60
  switchport mode access
!
```

SERVERROOM

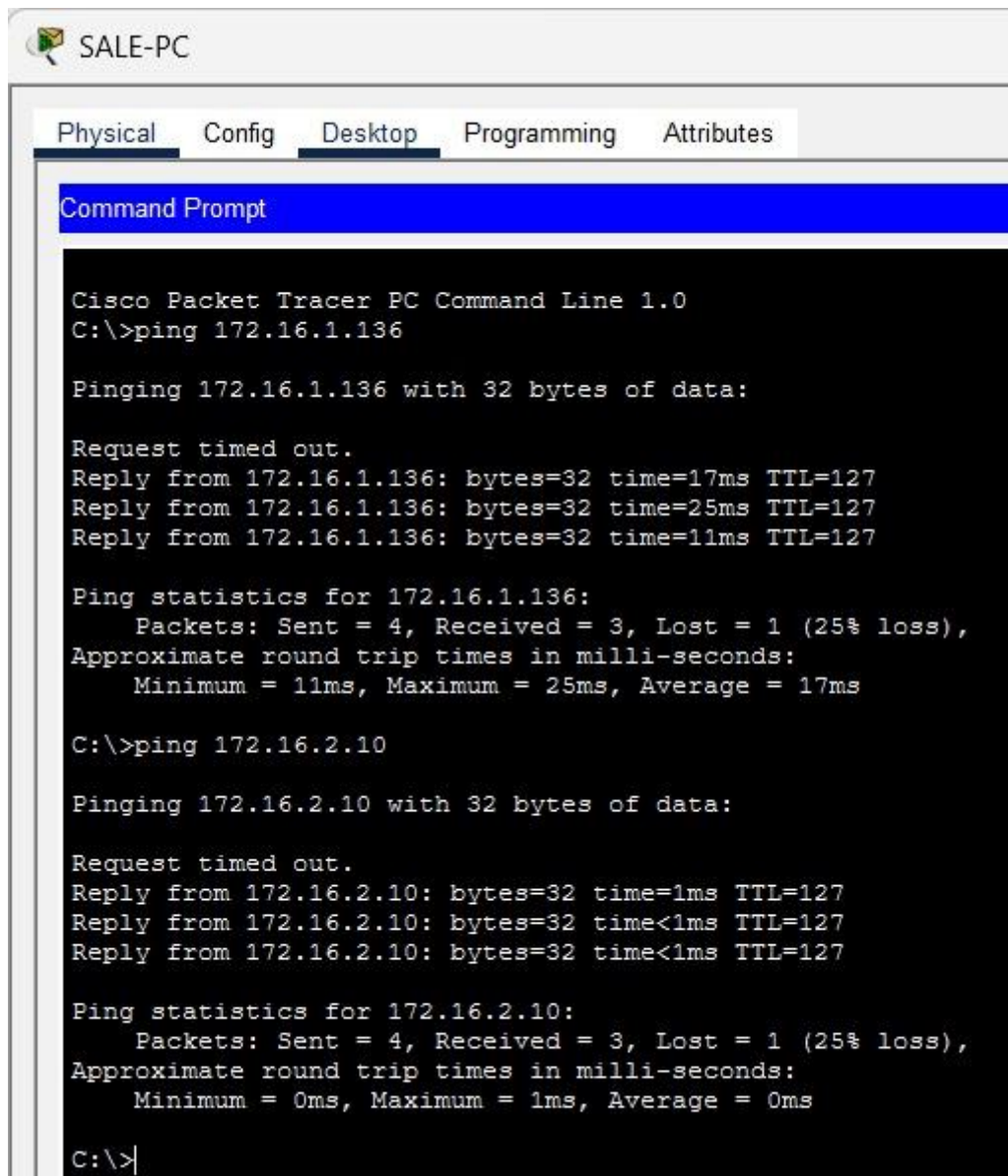
Physical Config CLI Attributes

```
interface FastEthernet0/23
  switchport access vlan 60
  switchport mode access
!
interface FastEthernet0/24
  switchport access vlan 60
  switchport mode access
!
interface GigabitEthernet0/1
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface GigabitEthernet0/2
  switchport access vlan 99
  switchport mode access
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
!
!
!
line con 0
!
line vty 0 4
  login
line vty 5 15
  login
!
!
!
!
end
```

Voila c'était tout en ce qui concerne les configurations maintenant nous allons passer au test de connectivités

TEST DE CONNECTIVITES :

Ping à partir de vlan 10



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named SALE-PC. The 'Desktop' tab is selected. The command prompt shows the execution of two ping commands. The first ping is to 172.16.1.136, which results in a 25% packet loss (1 out of 4 packets lost). The second ping is to 172.16.2.10, which results in a 25% packet loss (1 out of 4 packets lost).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.136

Pinging 172.16.1.136 with 32 bytes of data:

Request timed out.
Reply from 172.16.1.136: bytes=32 time=17ms TTL=127
Reply from 172.16.1.136: bytes=32 time=25ms TTL=127
Reply from 172.16.1.136: bytes=32 time=11ms TTL=127

Ping statistics for 172.16.1.136:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 25ms, Average = 17ms

C:\>ping 172.16.2.10

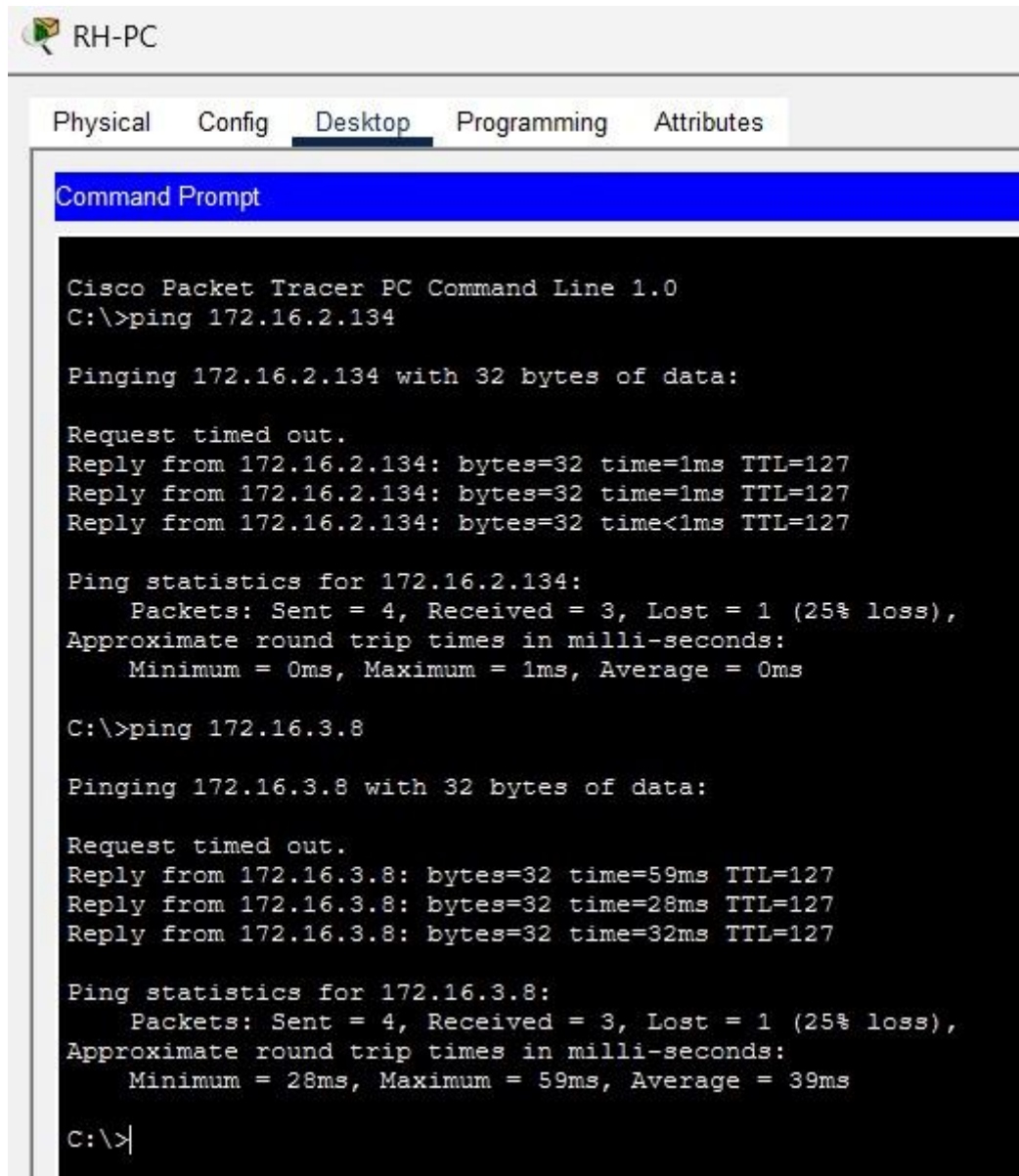
Pinging 172.16.2.10 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.10: bytes=32 time=1ms TTL=127
Reply from 172.16.2.10: bytes=32 time<1ms TTL=127
Reply from 172.16.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping à partir de vlan 20



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'RH-PC'. The 'Desktop' tab is selected. The command prompt shows the execution of two ping commands. The first command is 'ping 172.16.2.134', which results in a 25% packet loss (1 out of 4 packets lost). The second command is 'ping 172.16.3.8', which also results in a 25% packet loss (1 out of 4 packets lost). The output for both pings shows that 3 bytes were received out of 4 sent, with round trip times of 1ms and 39ms respectively.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.2.134

Pinging 172.16.2.134 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.134: bytes=32 time=1ms TTL=127
Reply from 172.16.2.134: bytes=32 time=1ms TTL=127
Reply from 172.16.2.134: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.2.134:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.16.3.8

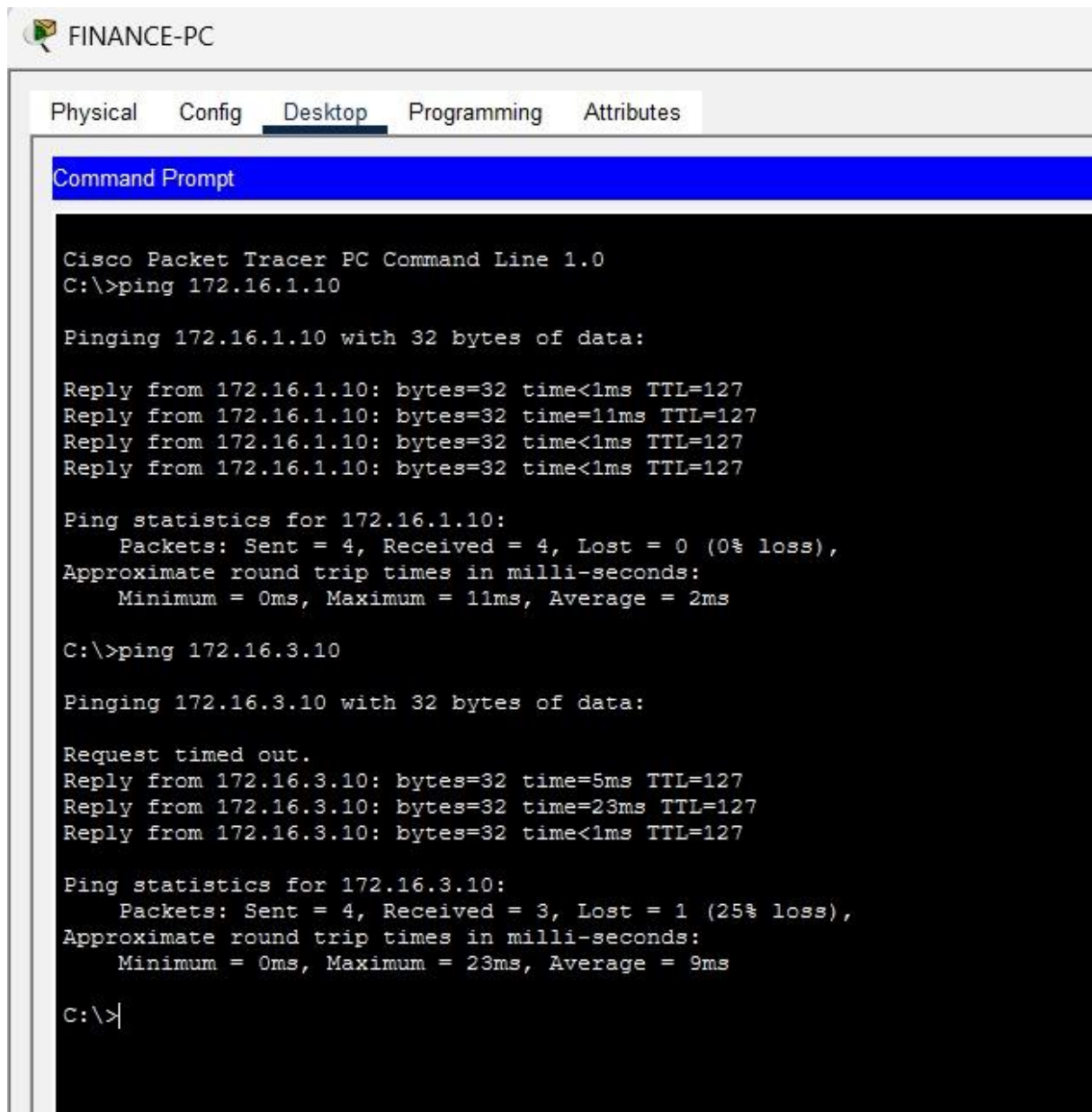
Pinging 172.16.3.8 with 32 bytes of data:

Request timed out.
Reply from 172.16.3.8: bytes=32 time=59ms TTL=127
Reply from 172.16.3.8: bytes=32 time=28ms TTL=127
Reply from 172.16.3.8: bytes=32 time=32ms TTL=127

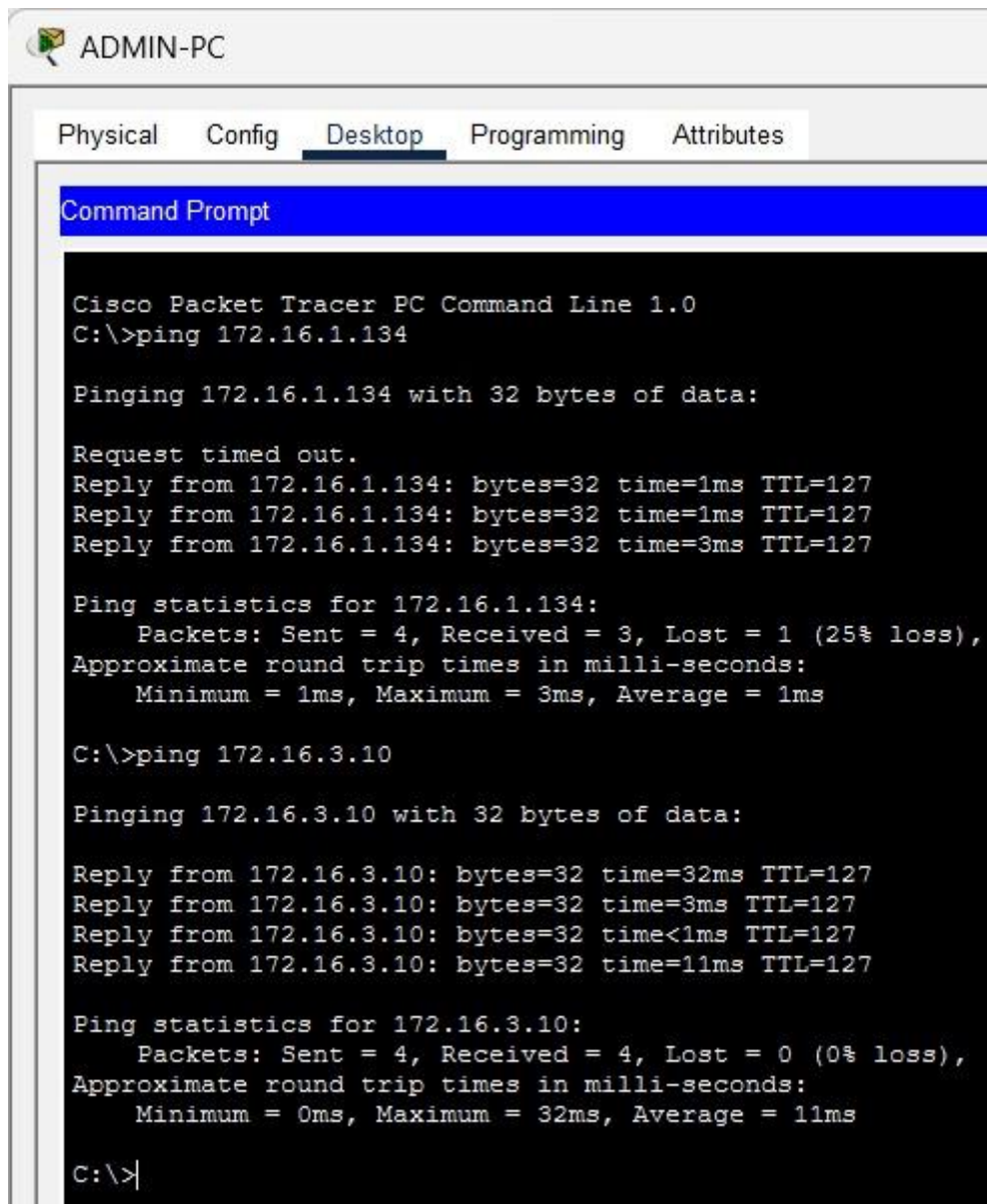
Ping statistics for 172.16.3.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 59ms, Average = 39ms

C:\>
```

Ping à partir vlan 30



Ping à partir de vlan 40



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'ADMIN-PC'. The 'Desktop' tab is selected. The command prompt shows the execution of two ping commands. The first ping is to 172.16.1.134, which results in a 25% packet loss (1 out of 4 packets lost). The second ping is to 172.16.3.10, which results in 0% packet loss (all 4 packets received).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.134

Pinging 172.16.1.134 with 32 bytes of data:

Request timed out.
Reply from 172.16.1.134: bytes=32 time=1ms TTL=127
Reply from 172.16.1.134: bytes=32 time=1ms TTL=127
Reply from 172.16.1.134: bytes=32 time=3ms TTL=127

Ping statistics for 172.16.1.134:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>ping 172.16.3.10

Pinging 172.16.3.10 with 32 bytes of data:

Reply from 172.16.3.10: bytes=32 time=32ms TTL=127
Reply from 172.16.3.10: bytes=32 time=3ms TTL=127
Reply from 172.16.3.10: bytes=32 time<1ms TTL=127
Reply from 172.16.3.10: bytes=32 time=11ms TTL=127

Ping statistics for 172.16.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 32ms, Average = 11ms

C:\>
```

Ping à partir de vlan 50

Command Prompt

```
Pinging 172.16.1.10 with 32 bytes of data:

Reply from 172.16.1.10: bytes=32 time<1ms TTL=127
Reply from 172.16.1.10: bytes=32 time=11ms TTL=127
Reply from 172.16.1.10: bytes=32 time<1ms TTL=127
Reply from 172.16.1.10: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 172.16.1.134

Pinging 172.16.1.134 with 32 bytes of data:

Reply from 172.16.1.134: bytes=32 time<1ms TTL=127
Reply from 172.16.1.134: bytes=32 time=10ms TTL=127
Reply from 172.16.1.134: bytes=32 time<1ms TTL=127
Reply from 172.16.1.134: bytes=32 time=3ms TTL=127

Ping statistics for 172.16.1.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms
```